原生 Python

Astra Automation

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原生 Python

开始之前

Python 是一种受欢迎的开发语言，尤其适用于数据中心自动化。在将 Python 的原生功能与多个通用软件包结合使用之前，您需要准备环境和所需的输入文件。

> 除了使用 Python 直接访问 Astra Control REST API 之外， NetApp 还提供了一个工具包软件包，用于抽象化 API 并消除一些复杂性。请参见 "NetApp Astra Control Python SDK" 有关详细信息 …

准备环境

下面介绍了运行 Python 脚本的基本配置要求。

您需要安装最新版本的 Python 3 。

必须安装 * 请求 * 和 * urllib3* 库。您可以根据环境需要使用 pip 或其他 Python 管理工具。

运行脚本的工作站必须能够访问网络并访问 Astra Control 。使用 Astra 控制服务时，您必须连接到互联网，并且能够连接到 `https://astra.netapp.io` 上的服务。

您需要一个具有帐户标识符和 API 令牌的有效 Astra 帐户。请参见 "获取 API 令牌" 有关详细信息 …

创建 JSON 输入文件

Python 脚本依赖于 JSON 输入文件中包含的配置信息。下面提供了示例文件。

> 您需要根据环境的具体情况更新这些示例。

以下文件包含 API 令牌和 Astra 帐户。您需要使用 ` -i` （或 ` -identity` ） CLI 参数将此文件传递到 Python 脚本。

```
{
  "api_token": "kH4CA_uVIa8q9UuPzhJaAHaGlaR7-no901DkkrVjIXk=",
  "account_id": "5131dfdf-03a4-5218-ad4b-fe84442b9786"
}
```

列出受管应用程序

您可以使用以下脚本列出 Astra 帐户的受管应用程序。

> 请参见 "开始之前" 所需 JSON 输入文件的示例。

```
#!/usr/bin/env python3
##-----------------------------------------------------------------------
------
#
# Usage: python3 list_man_apps.py -i identity_file.json
#
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#
##-----------------------------------------------------------------------
------

import argparse
import json
import requests
import urllib3
import sys

# Global variables
api_token = ""
account_id = ""

def get_managed_apps():
    ''' Get and print the list of managed apps '''

    # Global variables
    global api_token
    global account_id

    # Create an HTTP session
    sess1 = requests.Session()

    # Suppress SSL unsigned certificate warning
    urllib3.disable_warnings(urllib3.exceptions.InsecureRequestWarning)

    # Create URL
    url1 = "https://astra.netapp.io/accounts/" + account_id +
```

```
"/k8s/v1/managedApps"

    # Headers and response output
    req_headers  = {}
    resp_headers = {}
    resp_data    = {}

    # Prepare the request headers
    req_headers.clear
    req_headers['Authorization'] = "Bearer " + api_token
    req_headers['Content-Type'] = "application/astra-managedApp+json"
    req_headers['Accept'] = "application/astra-managedApp+json"

    # Make the REST call
    try:
        resp1 = sess1.request('get', url1, headers=req_headers,
allow_redirects=True, verify=False)

    except requests.exceptions.ConnectionError:
        print("Connection failed")
        sys.exit(1)

    # Retrieve the output
    http_code = resp1.status_code
    resp_headers = resp1.headers

    # Print the list of managed apps
    if resp1.ok:
        resp_data = json.loads(resp1.text)
        items = resp_data['items']
        for i in items:
            print(" ")
            print("Name: " + i['name'])
            print("ID: " + i['id'])
            print("State: " + i['state'])
    else:
        print("Failed with HTTP status code: " + str(http_code))

    print(" ")

    # Close the session
    sess1.close()

    return

def read_id_file(idf):
    ''' Read the identity file and save values '''
```

```
    # Global variables
    global api_token
    global account_id

    with open(idf) as f:
        data = json.load(f)

    api_token = data['api_token']
    account_id = data['account_id']

    return

def main(args):
    ''' Main top level function '''

    # Global variables
    global api_token
    global account_id

    # Retrieve name of JSON input file
    identity_file = args.id_file

    # Get token and account
    read_id_file(identity_file)

    # Issue REST call
    get_managed_apps()

    return

def parseArgs():
        ''' Parse the CLI input parameters '''

        parser = argparse.ArgumentParser(description='Astra REST API -
List the managed apps',
                                add_help = True)
        parser.add_argument("-i", "--identity", action="store", dest
="id_file", default=None,
                                help='(Req) Name of the identity input file',
required=True)

        return parser.parse_args()

if __name__ == '__main__':
    ''' Begin here '''
```

```
    # Parse input parameters
    args = parseArgs()

    # Call main function
    main(args)
```

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